



Module Presenter's Manual

For

Data Analysis with MS Excel

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Ver 1.0***

Amendment Record

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1. Introduction

At the end of this module, student will be able to:

- Explain customization of the Ribbon and Quick Access Toolbar
- Explain proofing and management of Add-ins
- Describe built-in and custom templates in Excel
- Explain importing and exporting of XML data in Excel
- Describe worksheet protection and locking/unlocking of cells
- Explain workbook protection, finalization, and encryption of workbook
- Explain shared workbook and change tracking in a shared workbook
- Explain the process of recovering unsaved versions of a file
- Explain tracing of formula precedents and dependants
- Explain Data Validation tool and checking common errors in a file
- Explain iterative calculation and setting calculation precision
- Describe the use of single cell and multi cell array formula
- Explain SUMIFS(), COUNTIFS(), and basic counting functions
- Describe Statistical and Financial functions
- Describe Date and Time functions
- Describe Text, Logical, and LookUp functions
- Explain the importance of chart and the different types of chart
- Explain modification of chart elements, chart area, plot area, and chart axes
- Describe formatting of chart title, legend, and gridlines
- Explain Trend lines and data series
- Explain Dual axes and chart templates
- Describe combination chart, error bars, and data tables
- Explain conditional formatting of data using data bars, color scales, and icon sets
- Explain Sparklines
- Describe sorting and filtering of labels and values in a PivotTable Report
- Describe the use slicer for a PivotTable
- Explain creation of PivotChart Report
- Explain changing of the chart type of a PivotChart Report
- Explain Analysis Tool Pak
- Explain single factor and two factor tool with and without replication
- Describe Correlation and Covariance
- Explain Exponential smoothing and F-Test two-sample for variances
- Describe Histogram and moving average
- Explain Random number generation and t-Test
- Describe Goal seek and Solver for what-if analysis
- Explain one-input and two-input data table and scenario
- Explain creating and running a macro
- Explain assigning macro to a command button
- Explain creation of a custom macro button on the Quick Access Toolbar
- Describe macro security
- Describe Form controls
- Work with Date, Time, Data Type and Conversion Functions
- Work with Financial Functions
- Link Workbooks and Summarize Data

2. Information on Session Allocation

Module	Online Hours	Self Study Hours
EXL	24	10

EXL: Data Analysis with MS Excel

3. Module Deliverables Available on OnlineVarsity

To aid the teaching process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description / Functionality
	Download Book - Student has the option to download the subject related e-book and read offline
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4	
1	Session 1 EXL – TL1	Session 2 EXL – TL2	Session 3 EXL – TL3	Session 4 EXL – TL4	S1-S3
2	Session 5 EXL – TL5	Session 6 EXL – TL6	Session 7 EXL – TL7	Session 8 EXL – TL8	
3	Session 9 EXL – TL9	Session 10 EXL – TL10	Session 11 EXL – TL11	Session 12 EXL – TL12	S4-S5

TL – Online Session

S – Self Study Session (over and above the module duration)

5. Session Coverage

The table below details out the contents of each session where:

Sess No.	Session Title	Session Details	Deliverables' Mapping
1 & 2	EXL – TL1 & TL2	All the topics as listed below from Session 1, Session 2, and Session 3 of <i>Data Analysis with MS Excel</i> book should be covered in this session. <u>Session 1 – Advanced Tools and Features in Excel</u> ➤ Understanding Add-ins ➤ Creating and Using Charts ➤ Working with Macros ➤ Linking Workbooks <u>Session 2 – Data Validation Using Functions</u> ➤ Built-in Data Validation Functions for Numbers and Text ➤ Customized Rules for Data Validation ➤ ISBLANK() ➤ ISNUMBER() ➤ ISTEXT() ➤ Lookup Functions with Data Validation <u>Session 3 – Performing Conditional Formatting</u> ➤ Built-in Rules to Highlight a Cell or a Cell Range ➤ Customized Conditional Formatting Rules ➤ Importing and Exporting XML Data	Data Analysis with MS Excel SG – Session 1,2, and 3 XP – Session 1,2, and 3 TG – Session 1, 2, and 3
3	EXL – TL3	The Try It Yourself questions from Session 1 to Session 3 of <i>Data Analysis with MS Excel</i> book should be covered in this session.	Data Analysis with MS Excel Session 1, 2, and 3
4	EXL –TL4	All the topics as listed below from Session 4 of <i>Data Analysis with MS Excel</i> book should be covered in this session. <u>Session 4 – Data Manipulation Using Functions</u> ➤ Functions to Rank, Sort, Filter, and Transpose data	Data Analysis with MS Excel SG – Session 4 XP – Session 4 TG – Session 4

Sess No.	Session Title	Session Details	Deliverables' Mapping
		➤ Conversion Functions ➤ TEXT() ➤ VALUE() ➤ DOLLAR() ➤ FIXED() ➤ DATEVALUE()	
	S1-S3	All the Assignments from Sessions 1 to 4 of Data Analysis with MS Excel course should be solved in these sessions.	Data Analysis with MS Excel (OnlineVarsity) Session 1-4
5	EXL –TL5	All the topics as listed below from Session 5 of Data Analysis with MS Excel book should be covered in this session. <u>Session 5 – Summarizing Data Using Functions</u> ➤ Mathematical and Trigonometric Functions ➤ Simplifying Calculations on Data ➤ Understanding Data Aggregation ➤ Summarizing Data with Database Functions ➤ DSUM() ➤ DMAX() ➤ DCOUNT() ➤ DAVERAGE()	Data Analysis with MS Excel SG – Session 5 XP – Session 5 TG – Session 5
6	EXL –TL6	The Try It Yourself questions from Session 4 and Session 5 of Data Analysis with MS Excel book should be covered in this session.	Data Analysis with MS Excel Session 4 and 5
7 & 8	EXL – TL7 & TL8	All the topics as listed below from Session 6 and Session 7 of Data Analysis with MS Excel book should be covered in this session. <u>Session 6 – Macros and Forms</u> ➤ Introduction to Pivot Table ➤ Working with Pivot Table ➤ Introduction to Pivot Chart ➤ Working with Pivot Chart <u>Session 7 – Data Analysis Using Statistical Functions</u> ➤ Moving Average for a Given Time Period	Data Analysis with MS Excel SG – Session 6 and 7 XP – Session 6 and 7 TG – Session 6 and 7

Sess No.	Session Title	Session Details	Deliverables' Mapping
		➤ Frequency Distribution for the Given Observations ➤ Hypothesis Test for a Sample Mean ➤ Analyzing Data Using Functions ➤ Sampling ➤ Covariance ➤ Correlation ➤ Goal Seek ➤ Solver ➤ F-test ➤ Histogram	
9	EXL – TL9	The Try It Yourself questions from Session 6 and Session 7 of <i>Data Analysis with MS Excel</i> book should be covered in this session.	Data Analysis with MS Excel Session 6 and 7
10 & 11	EXL – TL10 & TL11	All the topics as listed below from Session 8 of <i>Data Analysis with MS Excel</i> book should be covered in this session. <u>Session 8 – Working with Financial Functions</u> ➤ ANOVA ➤ Regression Analysis ➤ Statistical Functions ➤ SLOPE() ➤ RSQ() ➤ INTERCEPT() ➤ TREND() ➤ Designing a Dashboard ➤ Creating and Using Charts for Dashboard	Data Analysis with MS Excel SG – Session 8 XP – Session 8 TG – Session 8
12	EXL – TL12	The Try It Yourself questions from Session 8 of <i>Data Analysis with MS Excel</i> book should be covered in this session.	Data Analysis with MS Excel Session 8
	S4-S5	All the Assignments from Sessions 5 to 8 of <i>Data Analysis with MS Excel</i> course should be solved in these sessions.	Data Analysis with MS Excel (OnlineVarsity) Session 5-8

6. Library References

➤	Data Analysis with Microsoft Excel by Kenneth N. Berk
➤	Business Analysis with Microsoft Excel by Conrad Carlberg
➤	Microsoft Excel Data Analysis and Business Modeling by Curtis Frye

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